



Customer Shopping Behaviour Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions on spending patterns, customer segments, and subscription behaviour.

DATA ANALYSES PROJECT

PYTHON · SQL · POWER BI

Project Overview

The Goal

This project analyses customer shopping behaviour using transactional data to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour — all to guide strategic business decisions.

Dataset at a Glance

3,900

Rows (purchases)

18

Columns (features)

37

Missing values (Review Rating)

Dataset Features

The dataset spans three key dimensions of customer data, covering 18 columns across 3,900 purchase records.



Customer Demographics

Age, Gender, Location, and Subscription Status — the foundation for segmenting and profiling the customer base.



Purchase Details

Item Purchased, Category, Purchase Amount, Season, Size, and Color — capturing what customers bought and when.



Shopping Behaviour

Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, and Shipping Type — revealing how customers shop.

Exploratory Data Analysis in Python

We began with data preparation and cleaning in Python, establishing a reliable foundation for downstream analysis.

01

Data Loading & Initial Exploration

Imported the dataset using pandas. Used `df.info()` to check structure and `.describe()` for summary statistics.

02

Missing Data Handling

Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

03

Column Standardization

Renamed columns to snake_case for better readability and documentation.

04

Feature Engineering

Created `age_group` column by binning customer ages. Created `purchase_frequency_days` column from purchase data.

05

Data Consistency Check

Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

06

Database Integration

Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

SQL Analysis — Revenue & Discount Insights

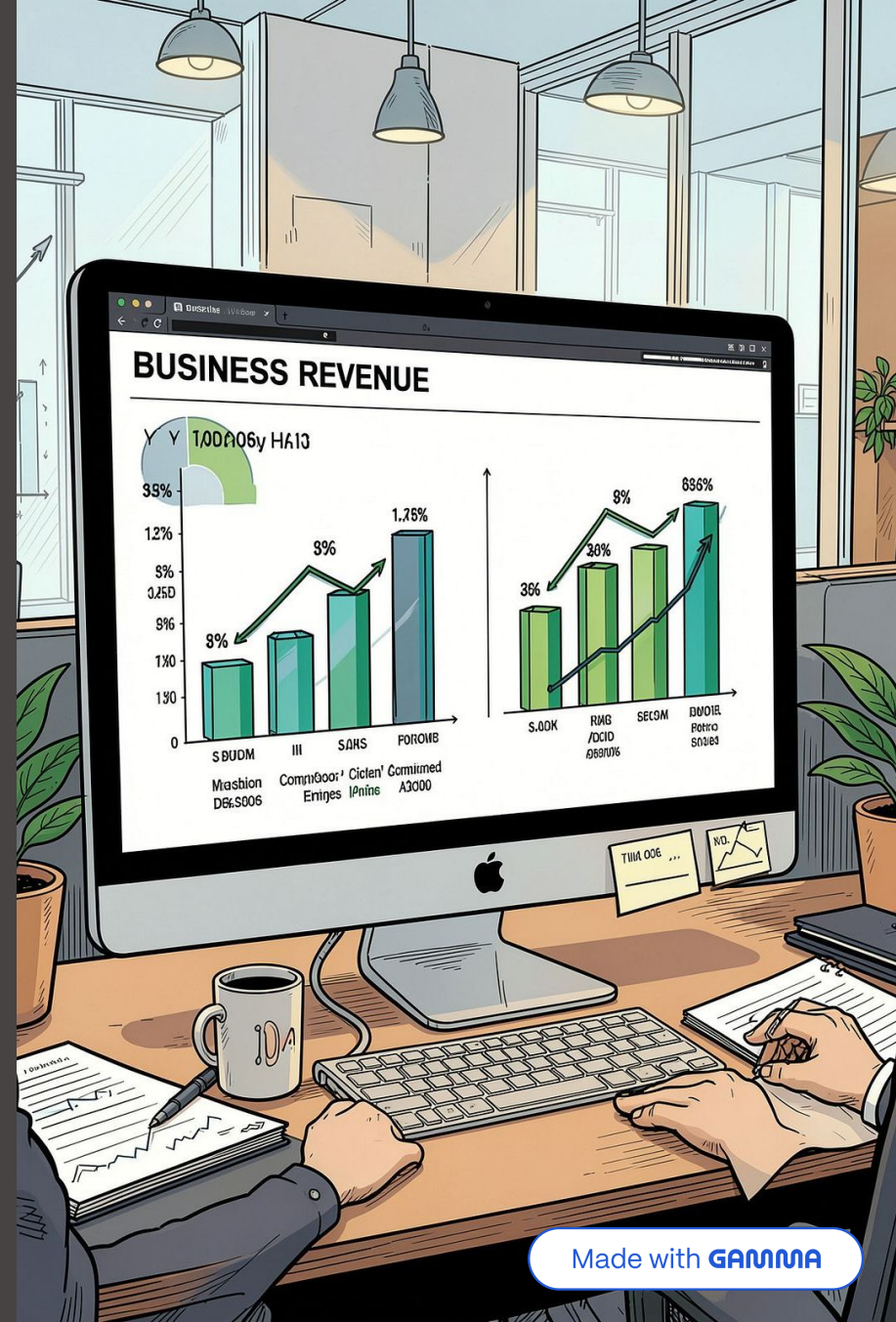
Structured analysis in PostgreSQL answered key business questions. First, revenue by gender was compared to understand spending differences between male and female customers.

Revenue by Gender

Compared total revenue generated by male vs. female customers to identify which segment drives more sales value.

High-Spending Discount Users

Identified customers who used discounts but still spent **above the average purchase amount** — revealing a segment that is price-sensitive yet high-value.



SQL Analysis — Products & Shipping

Top 5 Products by Rating

Found products with the highest average review ratings to identify customer favourites and quality leaders.

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping to assess whether shipping preference correlates with spend.

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status to quantify the value of subscriber relationships.

Discount-Dependent Products

Identified the 5 products with the highest percentage of discounted purchases, highlighting items most reliant on promotions to drive sales.

SQL Analysis — Customer Segmentation

Segmentation Logic

Customers were classified into three segments based on purchase history:

- **New** — First-time or infrequent buyers
- **Returning** — Customers with moderate repeat purchases
- **Loyal** — High-frequency, consistent buyers

Top 3 Products per Category

Listed the most purchased products within each category to understand category-level preferences and guide inventory and marketing focus.



SQL Analysis — Repeat Buyers & Age Groups

Repeat Buyers & Subscriptions

Checked whether customers with **more than 5 purchases** are more likely to subscribe — testing the hypothesis that purchase frequency drives subscription conversion.

Revenue by Age Group

Calculated total revenue contribution of each age group to identify which demographics deliver the highest business value and should be prioritized in marketing efforts.

Power BI Dashboard

Finally, we built an interactive dashboard in Power BI to present insights visually, consolidating findings from the Python and SQL analysis into a single interface for stakeholder review.



Business Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers to increase conversion and retention.



Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control, especially for discount-dependent products.



Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.



Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.